

Plan B-quarter — Revision notes (2026-04-24)

Derived from Plan A (docs/brd/zia-brd-2026-04-24.md, build-proven) after Plan B and Plan B-half both failed to build. Applies only 3 of 12 Plan B changes — the additive-text-only subset with no structural edits to types, picklists, or refs. Applied: C2 sanitize picklist parens/commas (with cross-ref sweep); C3 link_name callout blockquote; C5 file-upload directive note. Skipped: C1 strip currency, C4 timezone mandatory, C6 lookup displayformat, C7 split shared category picklist, C8 closed-picklist directive, C9 demote CRUD matrix, C10 compress workflows. If this builds, the breaker is among the skipped structural changes.

Expense Reimbursement Management — Business Requirements Document for Zoho Zia

Document ID: zia-brd-2026-04-24

Status: Phase 7 Final

§1 Ingestion Contract

```
doc_version: "1.0"
generated: "2026-04-24"
app_display_name: "Expense Reimbursement Management"
app_link_name: expense_reimbursement_management
currency: ZAR
timezone: Africa/Johannesburg
date_format: dd-MMM-yyyy
time_format: 24-hr
notation: see §3
how_to_read: >
  Reconstruct every artifact listed in §16; treat every table row
  as a schema directive.
```

Reconstruct every artifact listed in §16 using the form, field, report, role, workflow, and email-template tables in this document as schema directives.

§2 Application Meta

Property	Value
Display Name	Expense Reimbursement Management
Link Name	expense_reimbursement_management
Author	holger.gevers360
Version	1.0
Generated	06-Apr-2026 13:04:52
Currency	ZAR

Timezone	Africa/Johannesburg
Date Format	dd-MMM-yyyy
Time Format	24-hr
Forms	6
Fields	53
Lookups	5
Subforms	0
Reports	11 (9 list + 1 kanban + 1 pivottable)
Status Values	9
Roles	7 internal + 3 portal (combined row in §9.2)
Workflows	13
Custom APIs	4
Email Templates	18 (named) + 1 inline ad-hoc (see §15)
Widgets	0
Named Picklists	5

§3 Glossary & Notation Legend

Type notation

Symbol	Meaning	Example
lookup→<form>	Foreign-key lookup to named form	lookup→expense_claims
self_ref→<form>	Self-referential lookup	self_ref→expense_claims
fx:"<expr>"	Computed/formula field value	fx:"zoho.currenttime + 5y"
@Picklist:<name>	Reference to named picklist in §7	pick list @Picklist:category
pick{a,b,c}	Inline picklist (≤3 values, single-use)	pick list pick{Standard,High}
pick list (users-module)	Pick list bound to Zoho Users module	pick list (users-module)
pick list (self-ref)	Pick list referencing own form records	pick list (self-ref)

Flag string

Symbol	Meaning	Example
*	Required (must have)	flags cell: *

‡	Unique	flags cell: ‡
†	Private / hidden from default UI	flags cell: †
–	Empty / not applicable	any empty cell

CRUD cell alphabet

Symbol	Meaning	Example
C R U D	CRUD record-level access letters	CRU = create+read+update
^{1 2 3}	Scope qualifier footnote	R ¹ = read scoped per footnote 1

Reads / Writes micro-syntax

Reads: semi-separated tokens – input.<field>; zoho.<var>; <form>[<filter>].<field>
Multi-field pull: <form>[<filter>].(field1,field2,field3)
Writes: semi-separated assignments – input.<field> := <value>
Arrow shorthand also accepted: input.<field> → <value>

State machine transition row format

from_status | trigger | condition (guard) | to_status | actor_role | workflow_ref (→ WF-NN)

Email template contract

Subject column: literal template string with {{var}} tokens.
Variables column: comma-separated {{var}} names.
Body prose: omitted. intent column is one sentence only.

Field identifiers. The `Link_name` column in every §6 field table is the PROGRAMMATIC identifier (used in Deluge scripts, API calls, form references). This **MUST** be preserved exactly as written — do not rename, do not convert to snake_case differently, do not prepend/append tokens. The `Label` column is the human-readable display name **ONLY**. When `link_name` and `label` differ, always prefer `link_name` for any programmatic binding.

§4 Entity Catalog

form	label	kind	purpose	key_field	fk_to	
departments	Departments	ref	Lookup reference table of organizational departments.	department_id	—	
clients	Clients	ref	Lookup reference table of clients for	client_id	—	

			expense association.		
gl_accounts	GL Accounts	ref	Maps GL codes to expense categories, SARS provisions, and risk levels.	gl_code	—
approval_thresholds	Approval Thresholds	config	Configuration table defining tiered approval limits and dual-key rules.	tier_name	—
expense_claims	Expense Claims	txn	One record per employee expense reimbursement claim moving through multi-tier approval.	claim_id	departments, clients, gl_accounts, self (Parent_Claim_ID)
approval_history	Approval History	audit	Append-only audit log of every action taken on a claim during approval workflow.	claim	expense_claims

§5 Entity-Relationship Skeleton

from_form	field	to_form	cardinality	on_delete	self_ref
approval_history	claim	expense_claims	N:1	—	—
expense_claims	department	departments	N:1	—	—
expense_claims	client	clients	N:1	—	—
expense_claims	gl_code	gl_accounts	N:1	—	—
expense_claims	Parent_Claim_ID	expense_claims	N:1	—	yes

§6 Form Catalog

6.1 departments — Departments

Lookup reference table of organizational departments available for selection on expense claims.

link_name	label	type	flags	default	ref/pick	help
department_id	Department ID	auto number	—	autonumber(1)	—	Auto-generated department identifier.
name	Name	text	—	—	—	Department name as it appears in dropdown selections.
is_active	Active	boolean	—	true	—	Whether this department is currently active and available for selection.

Form-level rules: none.

6.2 clients — Clients

Lookup reference table of clients available for selection on expense claims.

link_name	label	type	flags	default	ref/pick	help
client_id	Client ID	auto number	—	autonumber(1)	—	Auto-generated client identifier.
name	Name	text	—	—	—	Client name as it appears in dropdown selections.
is_active	Active	boolean	—	true	—	Whether this client is currently active and available for selection.

Form-level rules: none.

6.3 gl_accounts — GL Accounts

Reference table mapping General Ledger account codes to expense categories, SARS provisions, and risk levels.

link_name	label	type	flags	default	ref/pick	help
gl_code	GL Code	text	—	—	—	The General Ledger account code (e.g., 6200). Must match your accounting system.

account_name	Account Name	text	—	—	—	Descriptive name for this GL account (e.g., Travel - Local Transport).
expense_category	Expense Category	pick list	—	—	@Picklist:category	The expense category this GL code maps to. Used for auto-assignment during approval.
receipt_required	Receipt Required	boolean	—	true	—	Whether a receipt/invoice is required for expenses in this category. Default: Yes (SARS S11(a)).
SARS_Provision	SARS Provision	text	—	—	—	Maps each GL code to SARS deduction provision, e.g. "S11(a) + S8(1)(a)".
Risk_Level	Risk Level	pick list	—	Standard	@Picklist:Risk_Level	ISO 37001: Categories prone to bribery risk receive enhanced scrutiny.
Active	Active	boolean	—	true	—	Whether this GL code is currently available for assignment.

Form-level rules: none. Referenced by WF-07, WF-09. ESG fields (ESG_Category, Carbon_Factor, GRI_Indicator) are NOT declared in this form — see §15 discrepancy 7.

6.4 approval_thresholds — Approval Thresholds

Configuration table defining tiered approval limits and dual-key authorization rules for the workflow.

link_name	label	type	flags	default	ref/pick	help
tier_name	Tier Name	text	—	—	—	Name of the approval tier (e.g., Tier 1 - Line Manager).
max_amount_zar	Max Amount ZAR	currency(ZAR)	—	—	—	Maximum claim amount (ZAR) this tier can approve; claims above escalate to next tier.
approver_role	Approver Role	text	—	—	—	The Zoho Creator role that approves claims at this tier.
Active	Active	boolean	—	true	—	Whether this threshold tier is currently active in the approval workflow.
Tier_Order	Tier Order	number	—	0	—	Numeric order in escalation chain; lower numbers approve first.
Requires_Dual_Approval	Requires Dual Approval	boolean	—	false	—	Whether this tier triggers dual-approval (two-key) authorization.
Dual_Approval_Role	Dual Approval Role	text	—	—	—	Role that acts as Key 2 approver when dual-approval is required.
Dual_Threshold_ZAR	Dual Threshold ZAR	currency(ZAR)	—	—	—	Amount threshold (ZAR) above which dual-approval is required.

Form-level rules: none (configuration form). Referenced by WF-07, WF-09, WF-13.

6.5 expense_claims — Expense Claims

Core transaction form; each record is one employee expense reimbursement claim moving through the multi-tier approval workflow.

link_name	label	type	flags	default	ref/1
Employee_Name1	Employee Name	name (compound)	*	—	pick{Mr.,Mrs.,I sub-compone
Email	Email	pick list (users- module)	*	—	users-module
Submission_Date	Submission Date	date time	*	—	—
claim_id	Claim ID	auto number	—	autonumber(1)	—
department	Department	lookup→departments	—	—	lookup→depa
Claim_Reference	Claim Reference	text	*	—	—
client	Client	lookup→clients	—	—	lookup→clien
Expense_Date	Expense Date	date	—	—	—
Department_Shadow	Department Shadow	text	†	—	—
category	Category	pick list	—	—	@Picklist:cate
Client_Shadow	Client Shadow	text	†	—	—

amount_zar	Amount ZAR	currency(ZAR)	—	—	—
Supporting_Documents	Supporting Documents	file upload	*	—	—
description	Description	rich text (textarea)	—	—	—
VAT_Invoice_Type	VAT Invoice Type	pick list	—	—	@Picklist:VAT_
POPIA_Consent	POPIA Consent	boolean	—	false	—
status	Status	pick list	—	—	@Picklist:statu
Rejection_Reason	Rejection Reason	rich text (textarea)	—	—	—

Version	Version	number	—	1	—
Retention_Expiry_Date	Retention Expiry Date	date	†	—	—
Parent_Claim_ID	Parent Claim ID	pick list (self-ref)	†	—	self_ref→expe
gl_code	GL Code	lookup→gl_accounts	—	—	lookup→gl_ac
Requires_Dual_Approval	Requires Dual Approval	boolean	†	false	—
Key_1_Approver	Key 1 Approver	text	†	—	—
Key_1_Timestamp	Key 1 Timestamp	date time	†	—	—
Key_2_Approver	Key 2 Approver	text	†	—	—
Key_2_Timestamp	Key 2 Timestamp	date time	†	—	—

Form-level rules:

1. BR-012: Expense_Date must not be in the future.
 2. BR-010: Expense_Date must not be older than 90 days.
 3. BR-011: amount_zar must be > 0.
 4. BR-006: Supporting_Documents must not be null (SARS S11(a) substantiation).
 5. BR-019: Duplicate detection — same employee, date, and amount triggers a warning.
 6. BR-007: Claims >= R5,000 require VAT_Invoice_Type = "Full Tax Invoice - R5000 and above".
 7. BR-008: POPIA_Consent must be true before submission.
-

6.6 approval_history — Approval History

Append-only audit log recording every action taken on an expense claim during the approval workflow.

link_name	label	type	flags	default	ref/pick	
claim	Claim	lookup→expense_claims	—	—	lookup→expense_claims	The claim audit relationship
action_1	Action	pick list	—	—	@Picklist:action_1	The take Sub App (LM, Reje Esca Res
actor	Actor	text	—	—	—	The or sy that perf the i
timestamp	Timestamp	date time	—	—	—	Date time actio reco
comments	Comments	rich text (textarea)	—	—	—	Data about actio incl amo code rejec reas

Form-level rules: none (audit form; append-only posture enforced at menu/report layer — see BR-023 and §16 #13).

§7 Picklist Appendix

action_1

Submitted
Submitted (Self-approval bypass)
Approved (LM)
Approved (HoD)
Approved (Key 1)
Approved (Key 2)

Rejected
Rejected (Key 2)
Reconsidered (Key 1)
Escalated (SLA Breach)
Resubmitted
Warning

Used by: `approval_history.action_1`

category

Travel
Accommodation
Meals and Entertainment
Office Supplies
Communication
Professional Services
Other

Used by: `expense_claims.category` ; `gl_accounts.expense_category`

Risk_Level

Standard
Elevated
High

Default: Standard

Used by: `gl_accounts.Risk_Level`

status

Draft
Submitted
Pending LM Approval
Pending HoD Approval
Pending Second Key
Key 2 Dispute
Approved
Rejected
Resubmitted

Used by: `expense_claims.status`

VAT_Invoice_Type

None
Abbreviated Tax Invoice - under R5000

Full Tax Invoice - R5000 and above

Used by: `expense_claims.VAT_Invoice_Type`

\$8 Reports Catalog

name	display_name	source_form	view_type	filter
expense_claims_Report	All Expense Claims	expense_claims	list	—
gl_accounts_Report	All GL Accounts	gl_accounts	list	—
gl_accounts_by_expense_category	Expense Categories	gl_accounts	kanban	—
approval_history_Report	All Approval Histories	approval_history	list	—
approval_thresholds_Report	All Approval Thresholds	approval_thresholds	list	—
clients_Report	All Clients	clients	list	—
departments_Report	All Departments	departments	list	—
Audit_Trail	Audit Trail	approval_history	list	—
pending_approvals_manager	Pending Approvals Manager	expense_claims	list	[status == "Pending LM Approval" status == "Pending HoD Approval"]
My_Claims	My Claims	expense_claims	list	[Added_User == zoho.loginuser]

Expense_Summary	Expense Summary	expense_claims	pivot	—

pending_approvals_manager columns (18 fields): Employee_Name1, Email, Submission_Date, claim_id, Claim_Reference, Expense_Date, department, client, category, amount_zar (total/avg/min/max), Supporting_Documents, description, status, Rejection_Reason, Version, Parent_Claim_ID, gl_code, Department_Shadow

My_Claims columns (17 fields): Employee_Name1, Email, Submission_Date, claim_id, Claim_Reference, Expense_Date, department, client, category, amount_zar, Supporting_Documents, description, status, Rejection_Reason, Version, Parent_Claim_ID, gl_code

§9 Roles & Access Model

9.1 Role Definitions

role	parent	kind	purpose	notes
Employee	—	internal	Base role for all staff submitting expense claims.	Self-approval prevention applies; can only view own claims.
Line Manager	Employee	internal	Stage 1 approval up to R999.99; SLA-bound.	Self-approval prevention: LM submitters bypass own tier.
Head of Department	Line Manager	internal	Stage 2 approval; Key 1 dual-approval authority.	Final authority under dual-threshold; oversees LM SLA compliance.
Finance Director	—	internal	Key 2 dual-approval authority for high-value claims.	Cannot be same person as Key 1 (King IV P7 segregation of duties).
Finance Accountant	—	internal	GL account management, exports, audit access.	Full GL CRUD; read-all on claims; no approval authority.
System Administrator	—	internal	Full system control across all forms.	Full CRUD + import/export everywhere.
Developer	—	internal	Development access for configuration and testing.	Read-only on all forms; no approval or admin authority.
Client Representative	—	portal	External client portal access to view claims.	View only; no edit or approval.

Vendor	—	portal	External vendor portal access to view claims.	View only; no edit or approval.
Customer	—	portal	Default portal profile.	Default portal permissions.

9.2 Record-Level CRUD Matrix

role \ form	departments	clients	gl_accounts	approval_thresholds	expense_claims	app
Employee	R	R	—	—	C R ¹ U ¹	R ¹
Line Manager	R	R	R	R	C R ² U ²	R ²
Head of Department	R	R	R	R	C R U ³	R ³
Finance Director	R	R	R	R	R U ⁴	R U
Finance Accountant	R	R	R	R	R U ⁵	R
System Administrator	CRUD	CRUD	CRUD	CRUD	CRUD	CRL
Developer	R	R	R	R	R	R
Portal (Client Representative / Vendor / Customer)	R	R	—	—	R ⁶	—

9.3 Scope Footnotes

- ¹ Employee can create and read/update only their own submitted claims (Added_User == zoho.loginuser).
- ² Line Manager can read/update claims in Pending LM Approval status assigned to their queue, plus own claims.
- ³ Head of Department can read/update all claims in Pending HoD Approval and Pending Second Key status.
- ⁴ Finance Director can read/update only claims in Pending Second Key status (Key 2 duty).
- ⁵ Finance Accountant can read/update GL account and threshold configuration; read all claims for audit.
- ⁶ Portal roles can read only claims associated with their client context; scope depends on Creator portal configuration.

9.4 Field-Level Override Table

form	field	Employee	Line Manager	HoD	Finance Director	Finance Accountant
expense_claims	status	R	R	R	R	C/R/U/D
expense_claims	gl_code	Hidden	R	R	R	C/R/U/D
expense_claims	Rejection_Reason	R	C/R/U	C/R/U	C/R/U	C/R/U/D
expense_claims	amount_zar	C/R/U	R	R	R	R
expense_claims	Key_1_Approver	Hidden	Hidden	Hidden	R	Hidden

expense_claims	Key_1_Timestamp	Hidden	Hidden	Hidden	R	Hidden
expense_claims	Requires_Dual_Approval	Hidden	Hidden	Hidden	R	Hidden
expense_claims	Department_Shadow	Hidden	Hidden	Hidden	Hidden	Hidden
expense_claims	Client_Shadow	Hidden	Hidden	Hidden	Hidden	Hidden
expense_claims	Parent_Claim_ID	Hidden	Hidden	Hidden	Hidden	Hidden

§10 Workflow State Machine

Status Enumeration

- **Draft** — pre-submission UI state; no workflow transitions lead back to Draft.
- **Submitted** — claim has passed on_validate; initial routing pending.
- **Pending LM Approval** — awaiting Line Manager tier-1 decision; SLA 3 days.
- **Pending HoD Approval** — escalated to Head of Department; either over tier-1 threshold or self-approval bypass.
- **Pending Second Key** — HoD (Key 1) approved; awaiting Finance Director (Key 2) dual-approval.
- **Key 2 Dispute** — Finance Director rejected; claim returned to HoD for reconsideration.
- **Approved** (*terminal*) — fully approved; no further transitions unless resubmission edit.
- **Rejected** (*terminal*) — rejected at LM or HoD tier; employee may edit and resubmit.
- **Resubmitted** (*re-entrant*) — employee has edited a rejected claim; re-enters routing cycle.

Transitions

from_status	trigger	condition	to_status	actor_role	workflow_ref
"Draft"	form_submit	POPIA consent = true; all validations pass	"Submitted"	Employee	→ WF-02
"Submitted"	form_submit	submitter holds Line Manager role	"Pending HoD Approval"	Employee	→ WF-03.B
"Submitted"	form_submit	submitter does not hold Line Manager role	"Pending LM Approval"	Employee	→ WF-03.A
"Pending LM Approval"	lm_approve	amount_zar ≤ tier-1 threshold	"Approved"	Line Manager	→ WF-07.A
"Pending LM Approval"	lm_approve	amount_zar > tier-1 threshold	"Pending HoD Approval"	Line Manager	→ WF-07.B
"Pending LM Approval"	lm_reject	always	"Rejected"	Line Manager	→ WF-08
"Pending LM Approval"	sla_timer	days since submission ≥ 3	"Pending HoD Approval"	SYSTEM	→ WF-13
"Pending HoD Approval"	hod_approve	amount_zar > dual-approval	"Pending Second Key"	Head of Department	→ WF-09.B

		threshold			
"Pending HoD Approval"	hod_approve	amount_zar ≤ dual-approval threshold	"Approved"	Head of Department	→ WF-09.B
"Pending HoD Approval"	hod_reject	always	"Rejected"	Head of Department	→ WF-10
"Pending Second Key"	fd_approve	approver ≠ Key_1_Approver	"Approved"	Finance Director	→ WF-11.B
"Pending Second Key"	fd_approve	approver == Key_1_Approver (same-person block)	"Pending Second Key"	Finance Director	→ WF-11.A
"Pending Second Key"	fd_reject	always	"Key 2 Dispute"	Finance Director	→ WF-12
"Key 2 Dispute"	hod_override	always	"Pending Second Key"	Head of Department	→ WF-09.A
"Rejected"	on_edit	employee edits and saves	"Resubmitted"	Employee	→ WF-06
"Resubmitted"	on_edit	resubmitter holds Line Manager role	"Pending HoD Approval"	Employee	→ WF-06
"Resubmitted"	on_edit	resubmitter does not hold Line Manager role	"Pending LM Approval"	Employee	→ WF-06

§11 Workflow Catalog

WF-01 expense_claim.on_load.auto_populate

- Trigger: expense_claims.on_load (Employee | new record only)
- Guard: form is in "add" mode
- Effect: Auto-fills Employee Name from zoho.loginuser and Email from zoho.loginuserid to reduce data-entry friction and ensure correct submitter association with the claim record.
- Reads: zoho.loginuser; zoho.loginuserid
- Writes: input.Employee_Name1 := zoho.loginuser; input.Email := zoho.loginuserid
- APIs: —
- Emails: —
- Audit: —

WF-02 expense_claim.on_validate

- Trigger: expense_claims.on_validate (Employee | form submission before save)
- Guard: always
- Effect: Enforces eight hard-stop validation gates: (1) Expense_Date not in future; (2) Expense_Date not older than 90 days; (3) amount_zar > 0; (4) Supporting_Documents not null; (5) duplicate detection by

date+amount+submitter; (6) claims >= R5,000 require Full Tax Invoice VAT type; (7) POPIA_Consent must be true. If all pass: sets status = "Submitted", records Submission_Date, calculates Retention_Expiry_Date = current timestamp + 5 years.

- Reads: input.Expense_Date; input.amount_zar; input.Supporting_Documents; input.VAT_Invoice_Type; input.POPIA_Consent; input.ID; zoho.currentdate; zoho.currenttime; zoho.loginuser; expense_claims[Expense_Date == input.Expense_Date && amount_zar == input.amount_zar && Added_User == zoho.loginuser && ID != input.ID]
 - Writes: input.status := "Submitted"; input.Submission_Date := zoho.currenttime; input.Retention_Expiry_Date := zoho.currenttime + 5 years
 - APIs: —
 - Emails: —
 - Audit: —
-

WF-03.A expense_claim.on_success — normal routing

- Trigger: expense_claims.on_success (Employee | after successful form submission)
 - Guard: submitter does NOT hold Line Manager role
 - Effect: Routes the new claim to a Line Manager for tier-1 approval. Creates an approval_history record logging the initial submission. Notifies the Line Manager by email with claim details (employee name, amount).
 - Reads: input.ID; input.claim_id; input.amount_zar; input.Employee_Name1; zoho.loginuser; zoho.currenttime; thisapp.permissions.isUserInRole("Line Manager")
 - Writes: input.status := "Pending LM Approval"
 - APIs: —
 - Emails: submit_notify_lm → line_manager
 - Audit: "Submitted"
-

WF-03.B expense_claim.on_success — self-approval bypass

- Trigger: expense_claims.on_success (Employee | after successful form submission)
 - Guard: submitter holds Line Manager role (self-approval prevention — King IV P1)
 - Effect: Bypasses the Line Manager tier and routes the claim directly to the Head of Department. Creates an approval_history record noting the self-approval bypass. Notifies the HoD by email.
 - Reads: input.ID; input.claim_id; input.amount_zar; input.Employee_Name1; zoho.loginuser; zoho.currenttime; thisapp.permissions.isUserInRole("Line Manager")
 - Writes: input.status := "Pending HoD Approval"
 - APIs: —
 - Emails: submit_self_approval_bypass → hod
 - Audit: "Submitted (Self-approval bypass)"
-

WF-04 expense_claim.on_success.fill_shadows

- Trigger: expense_claims.on_success (Employee | after successful form submission or edit)
- Guard: always
- Effect: Populates Department_Shadow and Client_Shadow with denormalized text copies of the linked department and client names respectively, to avoid nested lookups in reports and list views. Sets field to empty string if the linked record is null.
- Reads: input.department.name; input.client.name
- Writes: input.Department_Shadow := department name; input.Client_Shadow := client name
- APIs: —
- Emails: —

- Audit: —
-

WF-05 expense_claim.on_success.generate_ref

- Trigger: expense_claims.on_success (Employee | after successful form submission)
 - Guard: always
 - Effect: Auto-generates the human-friendly Claim_Reference in format "EXP-NNNN" by zero-padding the numeric claim_id to 4 digits (e.g., claim_id 5 → "EXP-0005"). This reference is used in all communications and notifications.
 - Reads: input.claim_id
 - Writes: input.Claim_Reference := "EXP-" + zero-padded(claim_id, 4)
 - APIs: —
 - Emails: —
 - Audit: —
-

WF-06 expense_claim.on_edit

- Trigger: expense_claims.on_edit (Employee | record edit/save on Rejected claim)
 - Guard: input.status == "Resubmitted"
 - Effect: Handles resubmission: increments Version, clears all dual-approval tracking fields (Key_1_Approver, Key_1_Timestamp, Key_2_Approver, Key_2_Timestamp, Requires_Dual_Approval), resets Submission_Date to now. Applies self-approval prevention: if resubmitter is Line Manager, routes to HoD; otherwise routes to Line Manager. Logs resubmission in approval_history and notifies the appropriate approver.
 - Reads: input.status; input.Version; input.ID; input.claim_id; zoho.loginuser; zoho.currenttime; thisapp.permissions.isUserInRole("Line Manager")
 - Writes: input.Version := input.Version + 1; input.Requires_Dual_Approval := false; input.Key_1_Approver := ""; input.Key_1_Timestamp := null; input.Key_2_Approver := ""; input.Key_2_Timestamp := null; input.Submission_Date := zoho.currenttime; input.status := "Pending HoD Approval" (if LM role) or "Pending LM Approval" (otherwise)
 - APIs: —
 - Emails: resubmit_self_approval_bypass → hod (if LM role); resubmit_notify_lm → line_manager (if not LM role)
 - Audit: "Resubmitted"
-

WF-07.A lm_approval.on_approve — under threshold (final approval)

- Trigger: expense_claims approval process > Line Manager (Level 1 on_approve)
- Guard: input.amount_zar ≤ tier-1 max_amount_zar from approval_thresholds (fallback R999.99)
- Effect: Grants final LM approval. Sets status to "Approved". Auto-populates gl_code and ESG metadata via lookup on gl_accounts by expense category. Logs audit trail with threshold detail and GL code. Notifies employee of approval.
- Reads: input.ID; input.claim_id; input.amount_zar; input.category; input.Email; zoho.loginuser; zoho.currenttime; approval_thresholds[tier_name == "Tier 1 - Line Manager"].(max_amount_zar); gl_accounts[expense_category == input.category].(gl_code, ESG_Category (advisory — not .ds-declared), Carbon_Factor (advisory — not .ds-declared))
- Writes: input.status := "Approved"; input.gl_code := GL account code from lookup; input.ESG_Category (advisory — not .ds-declared) := ESG classification; input.Estimated_Carbon_KG (advisory — not .ds-declared) := amount_zar × carbon_factor
- APIs: —
- Emails: lm_approved_final → input.Email
- Audit: "Approved (LM)"

WF-07.B lm_approval.on_approve — over threshold (escalate to HoD)

- Trigger: expense_claims approval process > Line Manager (Level 1 on_approve)
 - Guard: input.amount_zar > tier-1 max_amount_zar from approval_thresholds (fallback R999.99)
 - Effect: Escalates claim to HoD. Sets status to "Pending HoD Approval". Auto-populates gl_code and ESG metadata via lookup. Logs audit trail with threshold explanation. Notifies HoD of pending claim. Logs a Warning audit entry if threshold config is missing.
 - Reads: input.ID; input.claim_id; input.amount_zar; input.category; input.Email; zoho.loginuser; zoho.currenttime; approval_thresholds[tier_name == "Tier 1 - Line Manager"].(max_amount_zar); gl_accounts[expense_category == input.category].(gl_code, ESG_Category (advisory — not .ds-declared), Carbon_Factor (advisory — not .ds-declared))
 - Writes: input.status := "Pending HoD Approval"; input.gl_code := GL account code from lookup; input.ESG_Category (advisory — not .ds-declared) := ESG classification; input.Estimated_Carbon_KG (advisory — not .ds-declared) := amount_zar × carbon_factor
 - APIs: —
 - Emails: lm_approved_escalate → hod
 - Audit: "Approved (LM)"
-

WF-08 lm_approval.on_reject

- Trigger: expense_claims approval process > Line Manager (Level 1 on_reject)
 - Guard: always
 - Effect: Sets claim status to "Rejected". Logs rejection with reason in approval_history. Notifies the employee with the rejection reason and instructions to edit and resubmit.
 - Reads: input.ID; input.claim_id; input.amount_zar; input.Rejection_Reason; input.Email; zoho.loginuser; zoho.currenttime
 - Writes: input.status := "Rejected"
 - APIs: —
 - Emails: lm_rejected → input.Email
 - Audit: "Rejected"
-

WF-09.A hod_approval.on_approve — dispute reconsideration (Key 2 override)

- Trigger: expense_claims approval process > HoD (Level 1 on_approve)
 - Guard: input.status == "Key 2 Dispute" (HoD is overriding Finance Director's rejection)
 - Effect: HoD overrides the Finance Director's Key 2 rejection. Records Key 1 approval (Key_1_Approver, Key_1_Timestamp). Resets status to "Pending Second Key" for a new Key 2 cycle. Notifies the Finance Director of the HoD reconsideration. Logs audit with "Reconsidered (Key 1)".
 - Reads: input.status; input.ID; input.claim_id; input.amount_zar; input.Key_1_Approver; zoho.loginuser; zoho.currenttime
 - Writes: input.status := "Pending Second Key"; input.Key_1_Approver := zoho.loginuser; input.Key_1_Timestamp := zoho.currenttime
 - APIs: —
 - Emails: key1_reconsider_override → finance_director
 - Audit: "Reconsidered (Key 1)"
-

WF-09.B hod_approval.on_approve — normal approval

- Trigger: expense_claims approval process > HoD (Level 1 on_approve)
- Guard: input.status != "Key 2 Dispute" (standard HoD approval path)

- Effect: Reads dual-approval threshold from approval_thresholds. Auto-populates GL code and ESG metadata. If amount exceeds dual-threshold, records Key 1 approval and escalates to Finance Director (Pending Second Key). If under threshold, grants final approval (Approved) and notifies employee. In all paths logs full audit trail with GL and ESG context.
 - Reads: input.status; input.ID; input.claim_id; input.amount_zar; input.category; input.Email; zoho.loginuser; zoho.currenttime; approval_thresholds[Requires_Dual_Approval == true].(Dual_Threshold_ZAR) (fallback 5000.00); gl_accounts[expense_category == input.category].(gl_code, ESG_Category (advisory — not .ds-declared), Carbon_Factor (advisory — not .ds-declared), GRI_Indicator (advisory — not .ds-declared))
 - Writes: input.status := "Pending Second Key" (if over dual-threshold) or "Approved" (if under); input.Key_1_Approver := zoho.loginuser; input.Key_1_Timestamp := zoho.currenttime; input.Requires_Dual_Approval := true (if over dual-threshold); input.gl_code := GL code from lookup; input.ESG_Category (advisory — not .ds-declared) := ESG classification; input.Estimated_Carbon_KG (advisory — not .ds-declared) := amount_zar × carbon_factor
 - APIs: —
 - Emails: key1_approved_dual_required → finance_director (if amount > dual-threshold); hod_approved → input.Email (if amount ≤ dual-threshold)
 - Audit: "Approved (Key 1)" (if over-threshold); "Approved (HoD)" (if under-threshold)
-

WF-10 hod_approval.on_reject

- Trigger: expense_claims approval process > HoD (Level 1 on_reject)
 - Guard: always
 - Effect: Sets claim status to "Rejected". Logs rejection with reason in approval_history. Notifies the employee with rejection reason and instructions to edit and resubmit.
 - Reads: input.ID; input.claim_id; input.amount_zar; input.Rejection_Reason; input.Email; zoho.loginuser; zoho.currenttime
 - Writes: input.status := "Rejected"
 - APIs: —
 - Emails: hod_rejected → input.Email
 - Audit: "Rejected"
-

WF-11.A finance_approval.on_approve — same-person block

- Trigger: expense_claims approval process > Finance Director (Level 3 on_approve)
 - Guard: zoho.loginuser == input.Key_1_Approver (same-person conflict — King IV P7)
 - Effect: Blocks approval with a governance alert. Resets status to "Pending Second Key". Logs a "Warning" audit record. Sends an inline governance alert to the admin (zoho.adminuserid) noting the blocked same-person dual-key attempt.
 - Reads: input.Key_1_Approver; zoho.loginuser; input.ID; input.claim_id; input.amount_zar
 - Writes: input.status := "Pending Second Key"
 - APIs: —
 - Emails: (inline) Governance Alert → zoho.adminuserid [INLINE-01]
 - Audit: "Warning"
-

WF-11.B finance_approval.on_approve — Key 2 approval

- Trigger: expense_claims approval process > Finance Director (Level 3 on_approve)
- Guard: zoho.loginuser != input.Key_1_Approver (independence confirmed)
- Effect: Grants final dual-key approval. Populates Key_2_Approver and Key_2_Timestamp. Sets status to "Approved". Logs full approval_history entry with both Key 1 and Key 2 signatory details. Notifies employee of final dual-key approval.

- Reads: input.Key_1_Approver; zoho.loginuser; zoho.currenttime; input.ID; input.claim_id; input.amount_zar; input.Email
- Writes: input.status := "Approved"; input.Key_2_Approver := zoho.loginuser; input.Key_2_Timestamp := zoho.currenttime
- APIs: —
- Emails: key2_approved_final → input.Email
- Audit: "Approved (Key 2)"

WF-12 finance_approval.on_reject

- Trigger: expense_claims approval process > Finance Director (Level 3 on_reject)
- Guard: always
- Effect: Creates a Key 2 Dispute state; does NOT immediately reject the claim. Routes claim back to HoD for reconsideration. Finance Director's rejection reason is captured. Notifies both HoD (dispute details and instructions) and employee (generic under-review notice). Logs audit record with "Rejected (Key 2)" and dispute reason.
- Reads: input.ID; input.claim_id; input.amount_zar; input.Rejection_Reason; input.Key_1_Approver; input.Email; zoho.loginuser; zoho.currenttime
- Writes: input.status := "Key 2 Dispute"
- APIs: —
- Emails: key2_rejected_dispute → hod; key2_dispute_notify_employee → input.Email
- Audit: "Rejected (Key 2)"

WF-13 sla_enforcement_daily

- Trigger: sla_enforcement_daily (SYSTEM | scheduled daily)
- Guard: always (daily schedule; processes all qualifying claims each run)
- Effect: Monitors "Pending LM Approval" and "Pending Second Key" queues. For LM tier: if 2+ days since submission, sends reminder to LM (CC HoD); if 3+ days, auto-escalates to HoD (status → "Pending HoD Approval"), logs SLA Breach audit, notifies HoD. For Key 2 tier: if 2+ days since Key_1_Timestamp, sends reminder to Finance Director (CC admin); if 3+ days, escalates to admin/CEO for manual intervention with full context.
- Reads: expense_claims[status == "Pending LM Approval"].(Submission_Date, claim_id, amount_zar, Key_1_Approver, ID); expense_claims[status == "Pending Second Key"].(Key_1_Timestamp, claim_id, amount_zar, Key_1_Approver, ID); zoho.currentdate; zoho.currenttime; zoho.adminuser; zoho.adminuserid
- Writes: claim.status := "Pending HoD Approval" (for LM SLA 3-day breaches only)
- APIs: —
- Emails: sla_reminder → line_manager (if LM pending 2+ days); sla_escalation → hod (if LM pending 3+ days); key2_sla_reminder → finance_director (if Key 2 pending 2+ days); key2_sla_escalation → zoho.adminuserid (if Key 2 pending 3+ days)
- Audit: "Escalated (SLA Breach)" (for LM 3-day breaches; actor = SYSTEM)

\$12 Custom APIs

name	method	params	returns	
Get_Claim_Status	POST	claim_reference:Text(req)	status:Text; amount_zar:Decimal; department:Text; category:Text; submitted_date:Date; last_action:Text	-

Get_Dashboard_Summary	POST	department:Text(opt); date_from:Date(opt); date_to:Date(opt)	pending_count:Number; approved_count:Number; rejected_count:Number; total_amount_pending:Decimal; total_amount_approved:Decimal	—
Get_ESG_Summary	POST	date_from:Date(opt); date_to:Date(opt); department:Text(opt)	total_carbon_kg:Decimal; claim_count:Number	—
Get_SLA_Breaches	POST	threshold_days:Number(opt); department:Text(opt)	breached_count:Number; at_risk_count:Number	—

All 4 Custom APIs are externally invoked via REST; no internal Deluge workflow invokes a Custom API (all \$11 workflow APIs: lines are —).

\$13 Email Templates

id	trigger_workflow_ref	to	cc	subject_template
submit_notify_lm	WF-03.A	line_manager	—	Expense Claim {{claim_id}} requires your approval
submit_self_approval_bypass	WF-03.B	hod	—	Expense Claim {{claim_id}} - Direct HoD Review
resubmit_notify_lm	WF-06	line_manager	—	RESUBMITTED - Claim {{claim_id}} (v{{Version}})
resubmit_self_approval_bypass	WF-06	hod	—	RESUBMITTED - Claim {{claim_id}} (v{{Version}}) - Direct HoD Review
lm_approved_final	WF-07.A	input.Email	—	Expense Claim {{claim_id}} - Approved

lm_approved_escalate	WF-07.B	hod	—	Expense Claim {{claim_id}} - Requires HoD Approval
lm_rejected	WF-08	input.Email	—	Expense Claim {{claim_id}} - Rejected
hod_approved	WF-09.B	input.Email	—	Expense Claim {{claim_id}} - Final Approval
hod_rejected	WF-10	input.Email	—	Expense Claim {{claim_id}} - Rejected by HoD
sla_escalation	WF-13	hod	—	SLA BREACH - Claim {{claim_id}} escalated
sla_reminder	WF-13	line_manager	hod	REMINDER - Claim {{claim_id}} awaiting approval
key1_approved_dual_required	WF-09.B	finance_director	—	TWO-KEY APPROVAL - Claim {{claim_id}} requires your sign-off
key2_approved_final	WF-11.B	input.Email	—	Expense Claim {{claim_id}} - Final Approval (Two-Key)

key2_rejected_dispute	WF-12	hod	—	TWO-KEY DISPUTE - Claim {{claim_id}} rejected by Key 2
key2_dispute_notify_employee	WF-12	input.Email	—	Expense Claim {{claim_id}} - Under Review
key1_reconsider_override	WF-09.A	finance_director	—	TWO-KEY RECONSIDERATION - Claim {{claim_id}} re-approved by Key 1
key2_sla_reminder	WF-13	finance_director	admin	REMINDER (Key 2) - Claim {{claim_id}} awaiting approval
key2_sla_escalation	WF-13	zoho.adminuserid	—	SLA BREACH (Key 2) - Claim {{claim_id}} escalated
INLINE-01	WF-11.A	zoho.adminuserid	—	GOVERNANCE ALERT - Claim {{claim_id}} - Same- Person Key 2 Blocked

§14 Business Rules Index

BR-001 (governance) No user in the Line Manager role may approve their own submitted claim — enforced_by: WF-03.B, WF-06

BR-002 (governance) Line Manager submitters are automatically routed to HoD tier to prevent self-approval (King IV P1) — enforced_by: WF-03.B, WF-06

BR-003 (governance) All expense claim state transitions are recorded in approval_history with actor, timestamp, and context to ensure complete audit traceability — enforced_by: WF-03.A, WF-03.B, WF-06, WF-07.A, WF-07.B, WF-08, WF-09.A, WF-09.B, WF-10, WF-11.A, WF-11.B, WF-12, WF-13

BR-004 (governance) The approval_history form must be configured as append-only (no Edit/Delete/Duplicate at menu level) to prevent post-facto tampering — enforced_by: role_permission

BR-005 (governance) The Key 2 approver (Finance Director) must not be the same person as Key 1 approver (HoD); same-person attempt is blocked and logged (King IV P7) — enforced_by: WF-11.A

BR-006 (compliance) At least one supporting document (receipt or invoice) must be attached to satisfy SARS S11(a) substantiation requirements — enforced_by: WF-02 (form_validation)

BR-007 (compliance) Claims with amount_zar >= R5,000 require VAT_Invoice_Type = "Full Tax Invoice - R5000 and above" per the SARS VAT Act — enforced_by: WF-02 (form_validation)

BR-008 (compliance) POPIA_Consent must be true before claim submission is accepted — enforced_by: WF-02 (form_validation)

BR-009 (compliance) Retention_Expiry_Date must be set to Submission_Date + 5 years per SARS Tax Administration Act S29 — enforced_by: WF-02

BR-010 (compliance) Expense_Date must not be older than 90 days from current date at submission — enforced_by: WF-02 (form_validation)

BR-011 (financial) amount_zar must be greater than zero — enforced_by: WF-02 (form_validation)

BR-012 (financial) Expense_Date must not be in the future — enforced_by: WF-02 (form_validation)

BR-013 (financial) Claims within the tier-1 threshold (default R999.99) receive final approval from the Line Manager; claims above are escalated to HoD — enforced_by: WF-07.A, WF-07.B, config

BR-014 (financial) Claims exceeding the dual-approval threshold (default R5,000.00) require both Key 1 (HoD) and Key 2 (Finance Director) authorization — enforced_by: WF-09.B, config

BR-015 (financial) GL code is auto-assigned based on expense category lookup during LM or HoD approval; employees do not manually set GL code — enforced_by: WF-07.A, WF-07.B, WF-09.B, role_permission

BR-016 (sla) Claims awaiting LM approval for 2 days receive a reminder email to the LM (CC HoD) — enforced_by: WF-13

BR-017 (sla) Claims awaiting LM approval for 3 days are automatically escalated to HoD status (SLA auto-escalation) — enforced_by: WF-13

BR-018 (sla) Claims awaiting Key 2 (Finance Director) approval for 3 days escalate to admin/CEO for manual intervention — enforced_by: WF-13

BR-019 (data_integrity) Duplicate claims (same employee, expense date, and amount) trigger a warning at submission but do not block the claim — enforced_by: WF-02

BR-020 (data_integrity) Parent_Claim_ID is a self-referential pick list on expense_claims used to link resubmitted claims to their parent record; the self-ref edge must be configured as pick list filtered to expense_claims.ID —

enforced_by: WF-06, form_validation

BR-021 (data_integrity) Version number increments on every resubmission; dual-approval state is fully reset so the new version cycles through fresh approval routing — enforced_by: WF-06

BR-022 (compliance) ESG/carbon reporting is advisory only until the five referenced fields (gl_accounts.ESG_Category, gl_accounts.Carbon_Factor, gl_accounts.GRI_Indicator, expense_claims.ESG_Category, expense_claims.Estimated_Carbon_KG) are added to their respective forms; current workflow writes silently no-op — enforced_by: advisory (see §15)

BR-023 (ui) approval_history reports (approval_history_Report, Audit_Trail) must have menu permissions excluding Edit/Delete/Duplicate/Import/Export/Print to enforce append-only posture — enforced_by: role_permission

BR-024 (ui) Shadow fields (Department_Shadow, Client_Shadow), retention expiry, dual-key tracking fields, and Parent_Claim_ID must be hidden from all non-admin roles — enforced_by: role_permission

BR-025 (ui) External callers invoking Custom APIs must authenticate via Zoho OAuth; API permissions are declared in Creator's microservices layer — enforced_by: config

BR-026 (compliance) Organization-type and ESG toggle configuration (compliance_config) is advisory only until the compliance_config form is added to the .ds; dependent rules use config (compliance_config advisory) traceability tag — enforced_by: config (compliance_config advisory)

§15 Out of Scope / Known Discrepancies

1. [EXCLUSION] Widgets: 0 widgets are declared in forgedys.yaml; do NOT create any widget components.
2. [EXCLUSION] Subforms: 0 subform relationships exist in the .ds; do NOT add any subforms.
3. [EXCLUSION] Deluge source code: no Deluge script text is included in this BRD by design. Reconstruct from §11 plain-English behavioral specs.
4. [EXCLUSION] Portal role field-level CRUD: insufficient data to specify at field granularity; apply Creator default portal permissions.
5. [EXCLUSION] Seed data: approval_thresholds.json, clients.json, departments.json, gl_accounts.json reference data exists but is not embedded in this BRD. Load separately.
6. [DISCREPANCY] compliance_config form: field-descriptions YAML mentions compliance_config with 4 fields (Config_Key, Config_Value, Description, Active); this form is ABSENT from the .ds export. Do NOT create this form. Treat compliance_config-dependent rules as advisory only until a .ds export confirms the schema. Any §14 BR referencing organization-type/ESG toggles must tag enforced_by: config (compliance_config advisory) to preserve traceability (see BR-026).
7. [DISCREPANCY] Schema-workflow divergence: workflows WF-07 and WF-09 read gl_accounts.ESG_Category, gl_accounts.Carbon_Factor, and gl_accounts.GRI_Indicator, and WF-09 writes expense_claims.ESG_Category and expense_claims.Estimated_Carbon_KG — but NONE of these 5 fields are declared in the .ds schema for their respective forms. Treat these workflow reads/writes as advisory (they silently no-op in Creator until the fields are added). Do NOT add any of these 5 fields to §6.3 or §6.5.
8. [DISCREPANCY] Governance Alert email: WF-11.A contains an inline ad-hoc sendmail block (subject: "GOVERNANCE ALERT - Claim {claim_id} - Same-Person Key 2 Blocked") that is NOT a named template in email-templates.yaml. It is documented in §13 as INLINE-01 with id = INLINE-01 to distinguish it from the 18 named templates.

9. [DISCREPANCY] 3c-config.md summary states 19 email templates; authoritative count from email-templates.yaml is 18 named templates. The 19th was a miscounting artifact; INLINE-01 is the governance alert inline sendmail, not a 19th named template.
 10. [EXCLUSION] Pivot table internal autoview reference: Expense_Summary is built on "Autoview_1775232194200_expense_claims" — this is a Creator-internal artifact. Do NOT create it manually; Creator generates it automatically when the pivot report is configured.
 11. [EXCLUSION] Named types (0 declared), external integrations, Zoho Analytics connectors: none declared in forged.yaml; do NOT add them.
-

§16 Reconstruction Checklist

1. MUST have exactly 6 forms: departments, clients, gl_accounts, approval_thresholds, expense_claims, approval_history.
2. MUST have exactly 53 fields across all forms (departments: 3, clients: 3, gl_accounts: 7, approval_thresholds: 8, expense_claims: 27, approval_history: 5).
3. MUST have exactly 5 lookup edges (see §5 ER Skeleton), including 1 self-referential (expense_claims.Parent_Claim_ID → expense_claims).
4. MUST have 0 subforms.
5. MUST have exactly 11 reports: 9 list, 1 kanban (gl_accounts_by_expense_category), 1 pivot/summary (Expense_Summary).
6. MUST have exactly 9 expense_claims.status values: Draft, Submitted, Pending LM Approval, Pending HoD Approval, Pending Second Key, Key 2 Dispute, Approved, Rejected, Resubmitted.
7. MUST have exactly 5 named picklists (see §7): action_1, category, Risk_Level, status, VAT_Invoice_Type. (Employee_Name1_prefix is an inline sub-component, NOT a named picklist.)
8. MUST have exactly 7 internal roles: Employee, Line Manager, Head of Department, Finance Director, Finance Accountant, System Administrator, Developer. Plus 3 portal roles represented as 1 combined row in §9.2: Client Representative, Vendor, Customer.
9. MUST have exactly 13 Deluge workflows (WF-01 through WF-13); multi-branch scripts have sibling .A/.B sub-IDs but count as 1 workflow each. §11 contains exactly 17 H3 blocks.
10. MUST have exactly 4 Custom APIs: Get_Claim_Status, Get_Dashboard_Summary, Get_ESG_Summary, Get_SLA_Breaches.
11. MUST have exactly 18 named email templates PLUS 1 inline sendmail block (INLINE-01) in the finance_approval Level-3 on_approve actions block with subject "GOVERNANCE ALERT - Claim {claim_id} - Same-Person Key 2 Blocked". Verify template names match §11 Emails: references.
12. MUST have 0 widgets.
13. MUST configure approval_history reports (approval_history_Report, Audit_Trail) with menu permissions excluding Edit/Delete/Duplicate/Import/Export/Print; the form itself may retain a default Edit action button (menu-level enforcement suffices for append-only posture; the .ds form declares both on_add and on_edit action hooks).

14. MUST apply conditional formatting to expense_claims_Report status column using exact hex colors from .ds:
Approved = #1bbc9b, Rejected = #e84c3d, Pending states = #bd588b. Additional pending-variant colors:
#765f89, #107c91 as declared in .ds.
15. MUST auto-generate Claim_Reference in EXP-NNNN format via WF-05.
16. MUST set Retention_Expiry_Date = Submission_Date + 5 years via WF-02.
17. MUST have exactly 5 required (must have) fields — all on expense_claims: Employee_Name1, Email, Submission_Date, Claim_Reference, Supporting_Documents.
18. MUST have approval_history.action_1 picklist with exactly 12 values (in order): Submitted, Submitted (Self-approval bypass), Approved (LM), Approved (HoD), Approved (Key 1), Approved (Key 2), Rejected, Rejected (Key 2), Reconsidered (Key 1), Escalated (SLA Breach), Resubmitted, Warning.
19. MUST have CC recipients on exactly 2 named email templates: sla_reminder (cc = hod), key2_sla_reminder (cc = admin). All other templates cc = —.
20. The §9.2 CRUD matrix MUST have exactly 8 data rows (7 internal + 1 combined-portal) and exactly 6 form columns in dependency order.